
Population Pooling Confers Drift Resistance: Cross-Subject Decoders Age Slower Than Subject-Specific Ones in Mesoscale Cortex

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Abstract

1 Cortical representations *drift*: a decoder fit to one animal’s neural activity slowly
2 loses accuracy on that same animal over days, even when behavior is unchanged.
3 The standard remedy is periodic recalibration on fresh within-subject data. We
4 report a different form of robustness. On the open BraiDyn-BC cohort (25 mice,
5 cortex-wide widefield $\Delta F/F$ parcellated into 44 Allen-atlas regions, a ~ 15 -day
6 operant protocol), we decode four task events and compare, per mouse, how much
7 accuracy is lost across two weeks by (i) that mouse’s *own* early-week decoder and
8 (ii) a decoder *pooled* over the other 24 mice’s early-week data. The pooled de-
9 coder ages substantially less: the paired personal-minus-pooled aging asymmetry
10 is $+0.024$ AUC ($p = 2 \times 10^{-5}$, sign-flip; the own decoder ages more in 76%
11 of the 100 mouse–target pairs), and the effect holds for all four events individu-
12 ally ($p < 0.005$). A count-matched control is decisive about *why*: holding the
13 number of training events fixed, a decoder built from *other* individuals ages far
14 less than the subject’s own ($+0.023$ AUC for self vs. -0.006 for a single other
15 mouse at equal sample count), so the robustness is not a data-volume artifact but
16 follows from not overfitting the subject’s own drift-prone idiosyncrasies. Pooling
17 over *many* other individuals rather than one adds a further, smaller and direction-
18 consistent gain (to -0.012), yielding a graded self $\rightarrow$ one-other $\rightarrow$ many-others
19 ordering. Population pooling is thus a drift-mitigation strategy in its own right:
20 the small early-week accuracy cost of decoding across individuals is repaid as
21 temporal stability. Code and a streamed-feature pipeline are released.

22 1 Introduction

23 Representational drift—the gradual reorganization of a neural code over days while the animal’s
24 behavior stays fixed—is now documented across sensory, motor, and association cortex and in hip-
25 pocampus [14, 2, 16]. Its practical consequence for neural decoding is stark: a decoder trained on an
26 animal today degrades on that same animal weeks later. The dominant response is *recalibration*—
27 periodically refitting or adapting the decoder on fresh within-subject data [1, 4]—which requires
28 ongoing labelled data from the individual and treats each subject in isolation.

29 We ask whether the code’s cross-individual structure offers a complementary route to temporal ro-
30 bustness. Cortex-wide behavioral representations are known to be substantially shared across indi-
31 viduals [9, 6, 7], and recent models decode behavior zero-shot on unseen subjects [17]. That prior
32 work is evaluated at essentially one point in time. The question we pose is temporal: *does a decoder*
33 *built from a population of other individuals resist drift better than a decoder built from the subject*

34 *itself?* If individual brains drift along idiosyncratic, partly independent trajectories, then averaging
 35 over many individuals should cancel the individual-specific component of drift—so a pooled de-
 36 coder would age more slowly than a personal one. This is a concrete, testable prediction that, to our
 37 knowledge, has not been examined.

38 We test it on BraiDyn-BC [5], a widefield calcium-imaging resource whose ~ 15 -day operant proto-
 39 col and shared 44-region Allen parcellation make within-subject and cross-subject decoders directly
 40 comparable across time. Our contributions:

- 41 • **Pooling resists drift.** Across two weeks, a population-pooled decoder loses less accuracy
 42 than a mouse’s own decoder, for all four decoded events (paired asymmetry $+0.024$ AUC,
 43 $p = 2 \times 10^{-5}$; Section 4.1).
- 44 • **It is not a data-volume artifact.** A count-matched control shows that, at equal training-
 45 set size, other-individual data ages far less than the subject’s own; a graded self $\rightarrow$ one-
 46 other $\rightarrow$ many-others ordering isolates avoiding self-overfitting as the primary driver and
 47 cross-individual diversity as a secondary, consistent refinement (Sections 4.2–4.3).
- 48 • **A drift-mitigation framing.** The early-week accuracy penalty of cross-subject decoding
 49 is repaid as late-week stability, suggesting population priors as an alternative to per-subject
 50 recalibration. We release code and a no-raw-download streaming pipeline.

51 2 Related work

52 **Representational drift and its mitigation.** Drift is characterized within a single subject [14, 2,
 53 16, 13], and the engineering response is within-subject recalibration or latent re-alignment: mani-
 54 fold stabilization [1], supervised and self-calibrating recalibration [4, 12], adversarial and nonlinear
 55 latent-dynamics alignment to the subject’s own reference day [3, 11]. Every one of these stabilizes
 56 a decoder using *that subject’s* own later data; none uses a population of *other* subjects as the source
 57 of stability, which is our question.

58 **Cross-subject decoding.** Shared structure across individuals supports cross-subject readout via
 59 manifold alignment [8], contrastive embeddings [15], and, on this very cohort, atlas-aligned founda-
 60 tion models [17]. Pooling across participants is also known to *reduce overfitting*: HTNet trains one
 61 decoder on pooled multi-participant data precisely to “avoid overfit decoders” from single-subject
 62 data scarcity [10]. Two things separate that line from ours. First, the overfitting it avoids is *sample-*
 63 *size* overfitting (too little data per subject), whereas our count-matched control holds training-set size
 64 fixed and still finds the effect, so the overfitting we implicate is to the subject’s *drift-prone* features
 65 rather than to small samples. Second, and more basically, all of these target accuracy/transfer at a
 66 *fixed time*; the temporal-robustness question—whether the pooled decoder *ages* more slowly over
 67 days—is orthogonal and, to our knowledge, unaddressed.

68 **Conservation of cortex-wide dynamics.** That cortex-wide activity is dominated by shared, low-
 69 dimensional, movement-related components [9, 6, 7] is the premise that makes cross-subject pooling
 70 viable at all; it does not by itself imply anything about drift.

71 3 Data and methods

72 **Dataset.** BraiDyn-BC [5] (DANDI:001425, CC-BY): per mouse and session, hemodynamically
 73 corrected dorsal-cortex $\Delta F/F$ parcellated into 44 Allen Common Coordinate Framework regions
 74 at ~ 30 Hz, with time-aligned behavioral channels. Twenty-five mice each performed ~ 15 daily
 75 operant sessions over ~ 3 weeks. We stream only the 44-region traces and behavioral channels
 76 (never the raw movies).

77 **Features and events.** For each of four events (lever-pull, lick, reward, tone) we detect onsets as
 78 rising edges (1 s refractory) and form a 44-dimensional evoked feature: mean post-onset (0–1 s) par-
 79 cellated $\Delta F/F$ minus a pre-onset (−0.5–0 s) baseline. Negatives are matched-count windows drawn
 80 > 2 s from any event. The decoder is a standardized ℓ_2 -regularized logistic regression throughout.

81 **Temporal blocks.** Each mouse’s task sessions are pooled into an *early* block (days 1–5) and a *late*
 82 block (days 11–15). We report AUC in four regimes: within-mouse same-block (5-fold CV on
 83 early, WS), within-mouse across-block (train early $\rightarrow$ test late, WX), pooled same-block (train other

Table 1: Personal vs. pooled aging across days (AUC lost from early- to late-block test, holding training fixed), per event, $n = 25$ mice. Asymmetry = personal – pooled aging; positive $\Rightarrow$ the mouse’s own decoder ages more. p : paired sign-flip permutation.

Event	Personal aging	Pooled aging	Asymmetry (95% CI)	p	own ages more
Lever-pull	−0.012	−0.026	+0.014	0.002	68%
Lick	+0.018	−0.010	+0.028	0.0001	84%
Reward	+0.023	−0.010	+0.033	0.0001	80%
Tone	+0.013	−0.010	+0.023	0.005	72%
Pooled			+0.024	2×10^{-5}	76%

84 mice’s early $\rightarrow$ test held-out early, LS), and pooled across-block (train other mice’s early $\rightarrow$ test
85 held-out *late*, LX).

86 **Aging.** We define a decoder’s *aging* as its accuracy drop when the test set moves from the held-
87 out mouse’s early block to its late block, holding training fixed. Personal aging is $WS - WX$;
88 pooled aging is $LS - LX$. The per-mouse *asymmetry* is $(WS - WX) - (LS - LX)$; positive means
89 the own decoder ages more. Significance is a paired sign-flip permutation test over mice (20,000
90 permutations), with a cluster (over-mice) bootstrap 95% CI.

91 4 Results

92 4.1 A population-pooled decoder ages more slowly than a personal one

93 For every mouse we hold the test data fixed—that mouse’s own early then late block—and vary
94 only the decoder’s *training* source: the mouse itself versus the other 24. The pooled decoder loses
95 less accuracy across the two weeks (Table 1). The paired personal-minus-pooled aging asymmetry
96 is positive for all four events (lever-pull +0.014, lick +0.028, reward +0.033, tone +0.023), each
97 significant at $p < 0.005$; pooled over all 100 mouse–target pairs the asymmetry is +0.024 AUC
98 ($p = 2 \times 10^{-5}$), and the personal decoder ages more in 76% of pairs. The sign is consistent even
99 where both decoders happen to *improve* with practice (lever-pull): the pooled decoder improves
100 *more*. Read as a gap, the early-week advantage of the personal decoder over the pooled one shrinks
101 toward zero by the late week, because only the personal decoder ages.

102 4.2 The robustness is not a data-volume artifact

103 A pooled decoder is trained on $\sim 24\times$ more events than a personal one, so the obvious deflationary
104 explanation is that more data, not pooling, buys the robustness. We rule this out with a count-
105 matched control (Fig. 1, left). For each held-out mouse we fix the number of training events at
106 that mouse’s *own* early-block count n and compare three decoders on the same held-out early $\rightarrow$ late
107 aging: **self** (the mouse’s own n events), **one** (n events from a single other mouse), and **many** (n
108 events drawn across the other 24). The result is a clean ordering, consistent in all four events (means
109 over targets): self ages +0.023, one-other −0.006, many-others −0.012 AUC. Two things follow.
110 First, at *identical* sample count the subject’s own decoder ages +0.029 more than a single other
111 mouse’s—so volume is not the driver; training on data other than the subject’s own is. The subject’s
112 early-week decoder overfits idiosyncratic features that then drift, and any decoder that never saw
113 the subject sidesteps this. Second, pooling over many others rather than one ages a further −0.006
114 (many < one in 4/4 events), so cross-individual diversity contributes a smaller, secondary refinement
115 on top of the primary self-vs-other effect.

116 4.3 Diversity: a secondary scaling with the number of training individuals

117 Consistent with that secondary effect, sweeping the number of training mice $N \in \{1, 2, 4, 8, 16, 24\}$
118 (Fig. 1, right) shows pooled aging becoming mildly more favorable as N grows for lick (−0.001 $\rightarrow$
119 −0.010), reward (−0.004 $\rightarrow$ −0.010), and tone (+0.003 $\rightarrow$ −0.010); for lever-pull it saturates
120 immediately (≈ -0.025 from $N=1$). The scaling is modest (~ 0.01 AUC) next to the self-vs-other

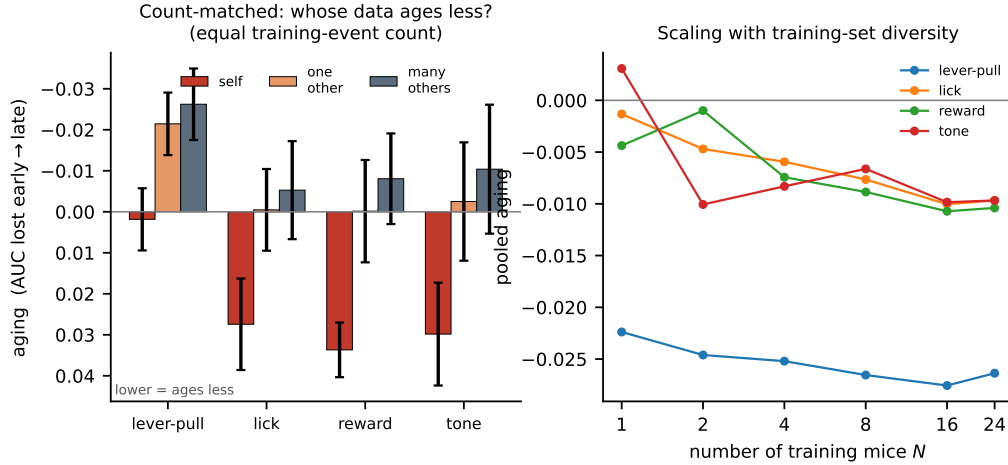


Figure 1: **Left:** count-matched control. At equal training-event count, decoder aging (AUC lost early→late; plotted so lower is more robust) for the subject’s own data (self) vs. one other mouse vs. many others. Self ages; other-individual data does not, even at matched count. **Right:** pooled aging vs. number of training mice N ; diversity adds a small further gain (saturating for lever-pull).

121 gap (~ 0.03), matching the count-matched decomposition: most of the drift resistance is already
 122 present with a single other individual, and additional individuals refine it.

123 5 Discussion

124 On a 25-mouse, 15-day widefield cohort, a decoder pooled over a population of other individuals
 125 resists cross-day drift better than a subject’s own decoder. A count-matched control shows this is
 126 not a data-volume artifact: at equal training-set size, other-individual data ages far less than the
 127 subject’s own, with cross-individual diversity adding a smaller, consistent refinement. The natural
 128 reading is that a subject’s own decoder overfits idiosyncratic, drift-prone features of that subject’s
 129 early sessions, while a decoder that never saw the subject can only latch onto the shared, temporally
 130 stable component of the code. This reframes population structure as a *temporal* resource: the well-
 131 known small accuracy penalty of cross-subject decoding is not merely tolerable but is repaid, over
 132 days, as stability. Where within-subject recalibration spends fresh labelled data from the individual
 133 to chase a drifting code, a population prior buys stability up front and for free at test time.

134 **Limitations.** The effect is measured at the mesoscale (parcel-averaged) level; whether it holds
 135 for single-neuron codes, where drift is characterized, is untested and is the natural next question.
 136 We study four thresholdable task events over a two-week window in one species; longer horizons,
 137 graded behaviors, and other modalities remain open. Our conclusions rest on within-vs-pooled and
 138 early-vs-late *comparisons* rather than absolute AUC. Finally, the count-matched control rules out
 139 data volume and points to self-overfitting, but the diversity increment is small and we do not identify
 140 which components of the drift are individual-specific.

141 Reproducibility

142 Data are open (BraiDyn-BC, DANDI:001425). We release the analysis code and a streaming feature
 143 pipeline that reproduces every number from the public parcellated traces without downloading raw
 144 imaging; all splits, permutation seeds, and hyperparameters are in Section 3 and the released code.

145 References

146 [1] A. D. Degenhart et al. Stabilization of a brain–computer interface via the alignment of low-dimensional
 147 spaces of neural activity. *Nature Biomedical Engineering*, 4:672–685, 2020.

148 [2] L. N. Driscoll et al. Dynamic reorganization of neuronal activity patterns in parietal cortex. *Cell*,
149 170(5):986–999, 2017.

150 [3] A. Farshchian et al. Adversarial domain adaptation for stable brain–machine interfaces. *ICLR*, 2019.

151 [4] B. Jarosiewicz et al. Virtual typing by people with tetraplegia using a self-calibrating intracortical brain–
152 computer interface. *Science Translational Medicine*, 7(313):313ra179, 2015.

153 [5] A. Kondo et al. BraiDyn-BC: a cortex-wide widefield calcium imaging dataset during a cued lever-pull
154 operant task. *Scientific Data*, 12, 2025. DANDI:001425.

155 [6] C. J. MacDowell and T. J. Buschman. Low-dimensional spatiotemporal dynamics underlie cortex-wide
156 neural activity. *Current Biology*, 30(14):2665–2680, 2020.

157 [7] C. J. MacDowell et al. Shared cortex-wide spatiotemporal motifs govern activity across individuals and
158 behavioral states. *Current Biology*, 34, 2024.

159 [8] S. Melbaum et al. Conserved structures of neural activity in sensorimotor cortex of freely moving rats
160 enable cross-subject decoding. *Nature Communications*, 13:7902, 2022.

161 [9] S. Musall et al. Single-trial neural dynamics are dominated by richly varied movements. *Nature Neuro-*
162 *science*, 22(10):1677–1686, 2019.

163 [10] S. M. Peterson, Z. Steine-Hanson, N. Davis, R. P. N. Rao, and B. W. Brunton. Generalized neural de-
164 coders for transfer learning across participants and recording modalities. *Journal of Neural Engineering*,
165 18:026014, 2021.

166 [11] B. M. Karpowicz, Y. H. Ali, L. N. Wimalasena, et al. Stabilizing brain–computer interfaces through
167 alignment of latent dynamics (NoMAD). *Nature Communications*, 16:4662, 2025.

168 [12] G. H. Wilson et al. Long-term unsupervised recalibration of cursor-based intracortical brain–computer
169 interfaces using a hidden Markov model. *Nature Biomedical Engineering*, 2025.

170 [13] M. E. Rule, T. O’Leary, and C. D. Harvey. Causes and consequences of representational drift. *Current*
171 *Opinion in Neurobiology*, 58:141–147, 2019.

172 [14] M. E. Rule et al. Stable task information from an unstable neural population. *eLife*, 9:e51121, 2020.

173 [15] S. Schneider, J. H. Lee, and M. W. Mathis. Learnable latent embeddings for joint behavioural and neural
174 analysis (CEBRA). *Nature*, 617:360–368, 2023.

175 [16] C. E. Schoonover et al. Representational drift in primary olfactory cortex. *Nature*, 594:541–546, 2021.

176 [17] M. Hosseini, E. Erturk, S. Hashemi, and M. M. Shanechi. Cross-subject modeling for widefield calcium
177 imaging via atlas-aligned spatiotemporal tokenization. In *Proceedings of the 43rd International Confer-*
178 *ence on Machine Learning (ICML)*, 2026. arXiv:2607.09754.